

# Effects of *Zostera marina* rhizosphere and leaf detritus on the concentration and distribution of pore-water sulfide in marine sediments

Alexandra G. Simpson<sup>a</sup>, Laura Tripp<sup>b</sup>, David H. Shull<sup>a,\*</sup>, Sylvia Yang<sup>b</sup>

<sup>a</sup> Department of Environmental Science, Western Washington University, Bellingham, WA 98225, USA

<sup>b</sup> Shannon Point Marine Center, 1900 Shannon Point Rd, Western Washington University, Anacortes, WA 98221, USA

## ARTICLE INFO

### Keywords:

Seagrass rhizosphere  
Seagrass detritus  
Sulfate reduction  
Sulfide  
Iron  
Diffusive gradients in a thin film (DGT)  
Sediment chemistry

## ABSTRACT

The relationship between seagrasses and pore-water hydrogen sulfide in the sediment is complex due to the opposing effects of the seagrass root system, which can reduce sulfide concentration by promoting its oxidation to sulfate, and the burial of organic matter from the plant itself, which can increase sulfide concentration by stimulating anaerobic sulfate reduction. Here, field samples of pore-water sulfide were collected in areas with and without *Zostera marina* shoots using Diffusive Gradients in Thin-Films (DGTs) to obtain two-dimensional visualizations of sulfide concentrations within the sediment in relation to location of eelgrass detritus and the rhizosphere. To decouple the effects of leaf detritus and the eelgrass rhizosphere on sulfide, manipulative laboratory studies were conducted over 4 weeks using 10 aquaria with or without eelgrass and different quantities of eelgrass detritus planted at 4 cm and 11 cm depths. Spatial distribution of pore-water sulfide was again measured via DGT. We found high spatial variation in pore-water sulfide concentration depending on the location analyzed relative to the rhizosphere or detritus. In the field study, pore-water sulfide concentration was greater in samples with eelgrass present, and the presence of leaf detritus accounted for higher than average sulfide concentrations in the sediment. In the laboratory experiment, sulfide concentrations localized around the leaf detritus increased with higher mass of added detritus, and sediment within the rhizosphere exhibited reduced sulfide concentrations. These results suggest that on the large scale the presence of eelgrass increases sulfide concentration, but on the smaller scale the pattern varies, with higher concentrations near buried detritus and lower concentrations near root tips.

## 1. Introduction

Pore-water hydrogen sulfide, a known toxicant inhibiting photosynthesis in seagrasses, works synergistically with other stressors such as low light availability to diminish plant growth and biomass and cause numerous other negative physiological effects in seagrass meadows (Bagarinao, 1992; Holmer and Bondgaard, 2001; Koch and Erksine, 2001; Pedersen et al., 2004; Borum et al., 2005, 2014; Korhonen et al., 2012; Dooley et al., 2013; Lamers et al., 2013). The accretion of plant detritus and other organic matter can cause seagrass beds to harbor higher levels of hydrogen sulfide in comparison to similar habitats lacking eelgrass (Harrison and Mann, 1975; Pollard and Moriarty, 1991; Isaksen and Finster, 1996; Holmer and Nielsen, 1997; Holmer et al., 2005) due to the production of sulfide through the anaerobic mineralization of sedimentary organic matter (Burdige, 2006; Mitsch and Gosselink, 2007).

But whether eelgrass will increase or decrease sulfide concentration might depend upon where in the sediment sulfide is measured. Whereas

the provision of organic detritus by eelgrass can lead to increased pore-water sulfide, diffusion of dissolved oxygen from the sediment surface or from eelgrass root tips can simultaneously decrease pore-water sulfide in nearby sediment by oxidizing it to sulfate (Frederiksen and Glud, 2006). Furthermore, iron reduction due to anaerobic decomposition of eelgrass organic matter produces Fe(II) which can reduce the concentration of free sulfide ( $\text{H}_2\text{S}$  and  $\text{HS}^-$ ) due to the reaction of sulfide with Fe(II) to produce insoluble iron sulfides (Jørgensen, 1977; Pagès et al., 2012).

However, few studies have detailed the multi-dimensional spatial gradients of sulfide and Fe(II) concentrations and the complexities of the root system (Pagès et al., 2012). It is possible to observe and quantify the interactions between eelgrass detritus, rhizosphere, sulfide, and Fe(II) concentrations by using Diffusive Gradients in Thin-Films (DGTs) and Diffusive Equilibration in Thin-Films (DETs). DGT/DETs have been used in seagrass sediments (Deborde et al., 2008; Cesbron et al., 2014; Pagès et al., 2012) as an *in-situ* method to quantify the distribution of various compounds including sulfide, Fe(II), trace

\* Corresponding author.

E-mail address: [david.shull@wwu.edu](mailto:david.shull@wwu.edu) (D.H. Shull).

metals, and nutrients within the seagrass rhizosphere.

The DGT/DET technique provides a two-dimensional visualization and quantification of sediment chemistry by utilizing and quantifying the diffusion of solutes into the sensor (Zhang and Davison, 1995). Solute transport is by molecular diffusion through a nitrocellulose filter membrane and a diffusive polyacrylamide hydrogel (Zhang and Davison, 1995; Robertson et al., 2008). Sulfide solutes bind to a resin gel and Fe(II) solutes are equilibrated in the diffusive gel before being exposed to ferrozine, which produces a magenta-colored complex (Zhang and Davison, 1995; Robertson et al., 2008). The DGT/DET technique allows sulfide to be measured at much higher spatial resolution than other conventional methods such as sediment cores or pore-water sippers. Furthermore, it is nearly impossible to know the proximity of these types of pore-water samplers to eelgrass root tips where sulfide intrusion can occur because it is not possible to observe the spatial relationship of a pore-water sipper to the location of the roots. Spatial resolution is important because low sulfide levels would likely be found near root tips whereas higher sulfide levels could be found further away. As such, DGT/DETs can show the spatial distribution of sulfide concentrations relative to eelgrass root tips and detritus.

The primary objective of this study was to determine how eelgrass detritus and eelgrass roots affect sulfide concentrations at different locations within the sediment. DGT/DETs were used in field and manipulative laboratory studies to measure the co-distribution of sulfide and Fe(II) in eelgrass-inhabited sediments at a 1-mm resolution.

## 2. Materials and methods

### 2.1. Description of study site

Located in the northwestern portion of Washington State, Padilla Bay contains over 3200 ha of eelgrass, a large percentage of the currently estimated 22,000 ha of *Z. marina* growing within the Salish Sea (Dowty et al., 2005; Nearshore Habitat Program, 2015). The field study was conducted in the southern portion of Padilla Bay near Bayview, WA during July 2015. Sediment, detritus, and live eelgrass shoots used in the laboratory experiment were collected from the same location as the field study during July 2015 and transported to Shannon Point Marine Center (Anacortes, WA). Although relatively low levels of sulfide (< 1 mM) are present in Padilla Bay, previous studies on eelgrass from Padilla Bay have shown sulfide levels as low as 1 mM can impact eelgrass physiology and growth (Walser, 2014).

### 2.2. DGT/DET preparation

To measure pore-water sulfide concentrations, DGTs utilize three layers: a nitrocellulose filter membrane (0.45-μm pore size; Bio-Rad), a diffusive polyacrylamide hydrogel and an AgI binding gel (Zhang and Davison, 1995; Rearick, 2004; Robertson et al., 2008). To measure Fe(II) concentrations, DETs are comprised of a polyacrylamide diffusive gel and nitrocellulose membrane which is placed against the sediment to equilibrate with pore water. After equilibration, the diffusive gel is exposed to another polyacrylamide gel, which has been equilibrated with a ferrozine solution. The AgI binding gels, diffusive polyacrylamide gels, and ferrozine equilibrated diffusive gels were constructed as detailed previously (Zhang and Davison, 1995; Rearick, 2004; Robertson et al., 2008). Milli-Q water (MQ; Millipore Element) was used to prepare all solutions and all components for gel construction were acid washed in 10% HCl prior to use. The diffusive gels had a final thickness of 0.76 mm and the binding gels had a final thickness of 0.5 mm. The DGT/DETs were placed in an N<sub>2</sub> chamber before use to remove possible artifacts from oxygen. After collection, DGT/DETs were scanned at 300 dpi using a flatbed scanner (Epson Workforce 325) in TIFF formats and analyzed via 8-bit grayscale color intensity using ImageJ version 1.49 (Rasband, 2015). Grayscale images were resized so

that 1 pixel was equivalent to 1 mm × 1 mm.

Sulfide and iron concentrations were obtained from the DGT/DETs following previously described methods (Robertson et al., 2008). To determine the sulfide concentration for the DGTs used in field and laboratory studies, a modified version of the standard curve was used to calculate theoretical sulfide uptake from grayscale intensities using Eq. (1):

$$U = \frac{I_g}{554.125 - 3.1789 \times I_g} \quad (1)$$

where U is the total uptake of sulfide by the gel (mol cm<sup>-2</sup>), I<sub>g</sub> is the grayscale intensity, and the numerical coefficients are from the standard curve (r<sup>2</sup> = 0.96). Once sulfide uptake was determined, a modified version of the DGT equation (Zhang and Davison, 1995; Robertson et al., 2008) was used to calculate actual sulfide concentrations:

$$C_{DGT} = \frac{U \Delta g}{Dt} \quad (2)$$

where C<sub>DGT</sub> is the sulfide concentration measured by the DGT, D is the diffusion coefficient (cm<sup>2</sup> sec<sup>-1</sup>), t is time (seconds), and Δg is the diffusive path length (cm). To determine Fe(II) concentrations for the DETs, the equation from a standard curve relating grayscale intensity to Fe(II) concentrations was used:

$$C_{DET} = \frac{I_g}{(0.633 - 0.0046 \times I_g)} \quad (3)$$

where C<sub>DET</sub> is the iron concentration measured by the DET (μM) and the numerical coefficients are from the standard curve (r<sup>2</sup> = 0.96).

### 2.3. DGT probe/aquaria assembly

To house the gels for the field study, multiple 17.8 cm × 12.1 cm sediment probes with 15.3 cm × 9.5 cm exposure windows were created using methods found in Robertson et al. (2008; Fig. 1). Similar housings were built to fit into an opening in the side of aquaria resembling “ant farms” (dimensions of 31 cm × 20 cm × 2.6 cm) filled with sediment from Padilla Bay (Fig. 1). An 18 × 12 cm opening in the side of each aquarium allowed the gel to be placed directly on the sediment and it could be removed without disrupting the sediment fabric. DGT/DETs for both field probes and aquaria gel holders were assembled with the AgI binding gel underlying the diffusive gel covered by a nitrocellulose filter membrane (0.45-μm pore size; Bio-Rad).

### 2.4. Field study

Eight field DGT probes were deployed during July 2015 at several locations within Padilla Bay, WA (locations centered at 48° 29' 35" N, 122° 29' 10" W) at low tide and they were collected before the flood tide covered the site, approximately four hours later. All DGT probes were deployed and collected within a 30-min timeframe to minimize any potential diel effects on sulfide concentration. Sites were selected based on proximity to eelgrass; 3 DGTs were deployed in bare spots and 5 DGTs were deployed adjacent to eelgrass rhizomes. Bare spots were considered “bare” only if no eelgrass shoots were observed in ca. 1-m radius from the location of the DGT probe. However, eelgrass roots and rhizomes might have been present in some of the “bare” areas. The probes were inserted into the sediment until the top of the DGT was 2–3 cm above the sediment surface. After collection, the probes were deconstructed and the binding gels were subsequently scanned between two cellulose acetate transparency sheets and analyzed following Robertson et al. (2008).

To quantify the distribution of roots and detritus in the sediment adjacent to the field-deployed DGTs, a vertical 2-cm thick slab of sediment adjacent to the DGT location was collected using a flat acrylic-coated shovel. Vertical and horizontal variations in sediment properties in this slab were examined by placing a 1.3 cm × 1.3 cm grid on top of

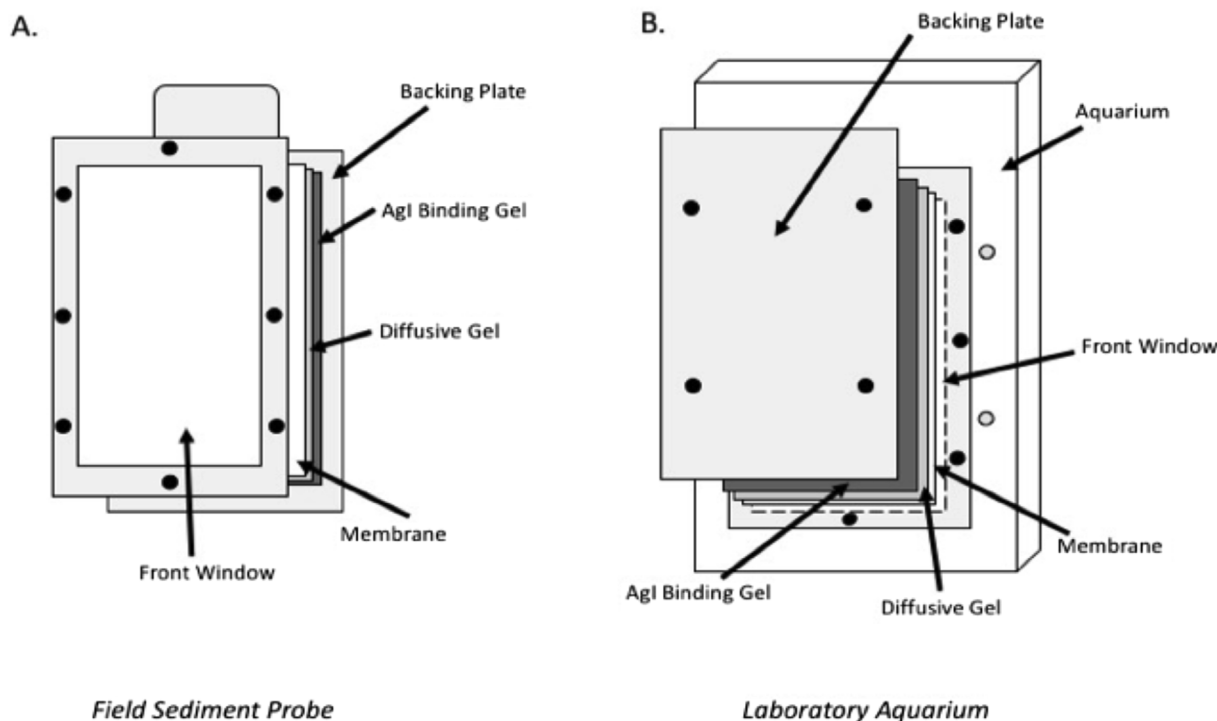


Fig. 1. Diagrams of field sediment probe (A) and laboratory aquarium detachable DGT (B).

the sediment slab and each grid cell was examined for roots and detritus as well as any other sources of sulfide production or reduction.

## 2.5. Laboratory experiment

To determine the mechanism through which eelgrass detritus and rhizospheres influence pore-water sulfide and iron concentration, ten aquaria were filled with Padilla Bay sediment obtained from “bare areas” and carefully excavated and placed into the aquarium to preserve the vertical stratigraphy. Detritus pieces measuring 7.6 cm in length and an average organic matter content of 0.0103 g (ash-free dry weight) were cut from the oldest leaves of nearby large eelgrass shoots. Eelgrass leaf detritus of varying amounts (0, 1, 2, 4, or 8 leaf sections resulting in a range of 0–0.082 g of added organic matter) was added to the sediment on one (randomly selected) side of ten aquaria at depths of 4 cm and 11 cm (Fig. 2). Live eelgrass shoots with an average length of 20 cm were transplanted with one shoot in each of five aquaria. Aquaria were placed in outdoor tanks with flowing seawater at Shannon Point Marine Center to allow for eelgrass to grow under natural light conditions. In order to block sediment from sun exposure, aquaria were covered in black plastic while ensuring eelgrass shoots had access to light. Eelgrass growth rates were monitored using the pin-prick method (Short and Duarte, 2001).

DGTs and DETs were deployed in each tank for four-hour periods once a week over four weeks to determine how sulfide and iron concentrations varied over time. DGTs and DETs were deployed and collected in the same order within approximately 30 min of each other each week to ensure that they were deployed for equal lengths of time. After collection, the DGTs and DETs were deconstructed. The binding gels of the DGTs and the ferrozine-equilibrated diffusive gels of the DETs were subsequently scanned between transparency sheets and analyzed.

As with the field study sediment slabs, sediment in all aquaria was analyzed using a 1.3 cm × 1.3 cm grid to search for roots and detritus. Grid cell contents were compared with sulfide and iron concentration as determined from the DGTs and DETs to determine if sulfide or iron concentration and distribution were influenced by varying levels of

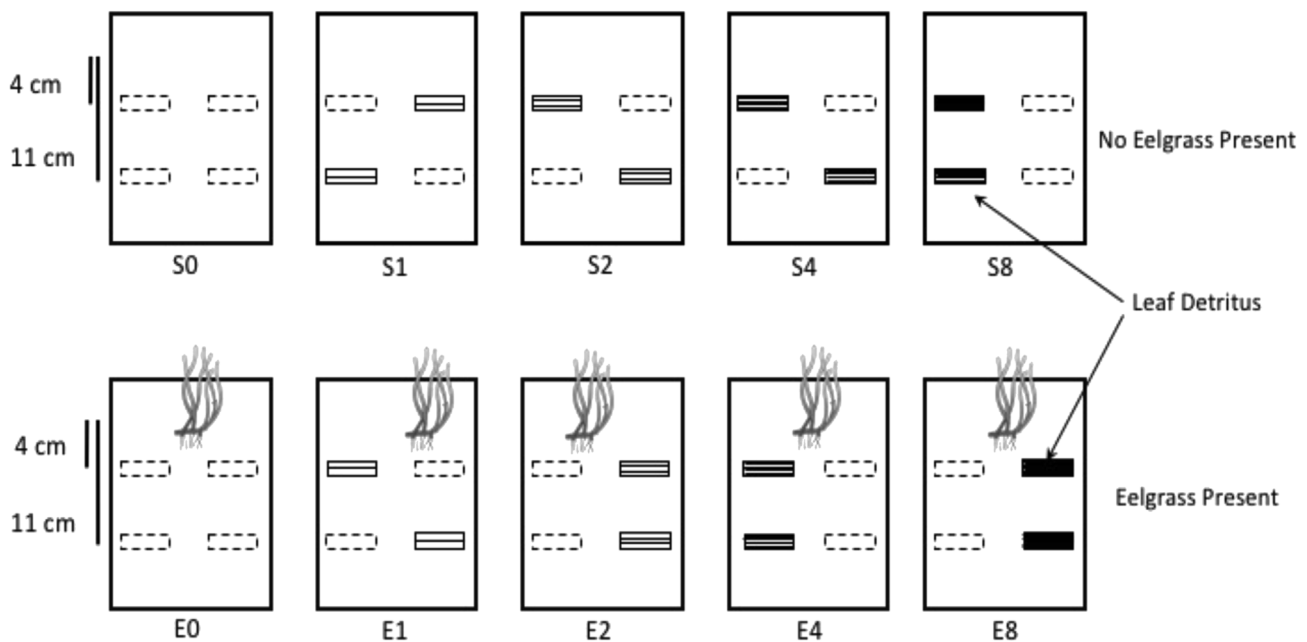
organic detritus.

## 2.6. Statistical analysis

For the field experiment, analyses of sulfide concentrations were based on three comparisons: (1) no eelgrass shoots vs eelgrass shoots; (2) detritus vs no detritus; and, (3) root zone vs no root zone. The mean sulfide concentration of the entire gel was used to determine sulfide concentration differences between no eelgrass shoot and eelgrass shoot locations using one-way analysis of variance (ANOVA) (to account for unequal sample size). Detritus and root zones were identified by examination of the sediment slab adjacent to the DGT/DET. A paired *t*-test was used to analyze the effects of detritus or roots on sulfide concentration.

For the laboratory experiment, analyses of sulfide concentrations were based on four comparisons: (1) eelgrass shoot vs no eelgrass shoot; (2) detritus vs no detritus; (3) root zone vs no root zone; and, (4) detritus at 4 cm vs detritus at 11 cm. Detritus and root zones were defined by the presence of either the added detritus or the presence of eelgrass roots observed in the sediment at the end of the laboratory experiment. Analysis of co-variance (ANCOVA) was used to compare the mean sulfide and Fe(II) concentration of the entire gel between eelgrass presence and no eelgrass treatments, where the amount of added detritus was considered a covariate. A paired *t*-test was used to compare sulfide and Fe(II) concentrations between the root zone and non-root zone and between areas with detritus and without detritus. An *ad hoc* pairwise *t*-test determined which detritus additions differed. To quantify the influence of depth on sulfide and Fe(II) concentration, the horizontal concentration mean at each depth of the gel, including the areas where detritus was added, in each aquarium was calculated. All statistical analyses were conducted using R statistical software (R Core Team, 2016).

At the end of the aquarium experiment, large numbers of shell fragments were found in the eelgrass aquarium with 1 detritus leaf (E1). The shell likely influenced sediment chemistry by limiting the flow of pore-water throughout the sediment. Therefore, data were statistically analyzed both with and without this aquarium and E1 was not included



**Fig. 2.** Diagram of the laboratory experiment. “S” indicates tanks with sediment only. “E” indicates tanks with live eelgrass. E0 and S0 had 0 detritus leaves added, E1 and S1 had 1, E2 and S2 had 2, E4 and S4 had 4 and E8 and S8 had 8. Detritus was located 4 cm and 11 cm from the sediment surface. Filled blocks indicate where leaf detritus was added and the dashed block indicates the corresponding area used in statistical analyses. Eelgrass illustration courtesy Diana Kleine, Integration and Application Network (IAN), University of Maryland Center for Environmental Science ([ian.umces.edu/imagelibrary/](http://ian.umces.edu/imagelibrary/)).

in the standard curve creation. Although we followed the development of slowly increasing sulfide concentrations in the aquaria for four weeks, we selected week two for detailed statistical analysis. This is because by week two sufficient sulfide had built up in the aquaria to enable us to assess how eelgrass shoots and detritus affected sulfide concentrations but sulfide forming around buried detritus had not yet bled into adjacent regions in the aquaria.

### 3. Results

#### 3.1. Field study

DGT probes adjacent to eelgrass shoots yielded slightly higher average sulfide concentrations ( $0.047 \text{ mM} \pm 0.020$  95% C.I.) compared to those placed in bare patches ( $0.028 \text{ mM} \pm 0.004$  95% C.I., Fig. 3). And, for probes placed next to eelgrass shoots, sulfide concentrations were higher near buried eelgrass leaf detritus compared to the average sulfide concentration measured by each corresponding probe (paired *t*-test,  $t = -13.9$ ,  $df = 7$ ,  $p < 0.001$ , Fig. 4). Interestingly, this pattern was not detected in probes placed in bare patches. Detritus was sparingly found in these areas and it appeared to have little influence on sulfide concentrations (Fig. 3). Although the presence of a root zone was not associated with significantly lower than average sulfide concentrations in eelgrass-adjacent field probes (Fig. 4; paired *t*-test,  $t = -1.63$ ,  $df = 7$ ,  $p = 0.15$ ), the root zone areas of the eelgrass-adjacent probes contained an average sulfide concentration of  $0.036 \text{ mM}$ , nearly  $0.012 \text{ mM}$  lower than the overall mean sulfide concentration (Fig. 4). Furthermore, root zones of sulfide probes possessed lower sulfide concentrations than detritus patches imaged by those same probes (Fig. 4; paired *t*-test,  $t = -13.8$ ,  $df = 7$ ,  $p < 0.001$ ). Overall, the presence of root zones and leaf detritus had opposite effects on sulfide concentration in adjacent sediment.

#### 3.2. Laboratory experiment: presence of live eelgrass shoots

To further investigate the roles of roots and eelgrass leaf detritus in determining sediment sulfide distribution and concentration, we

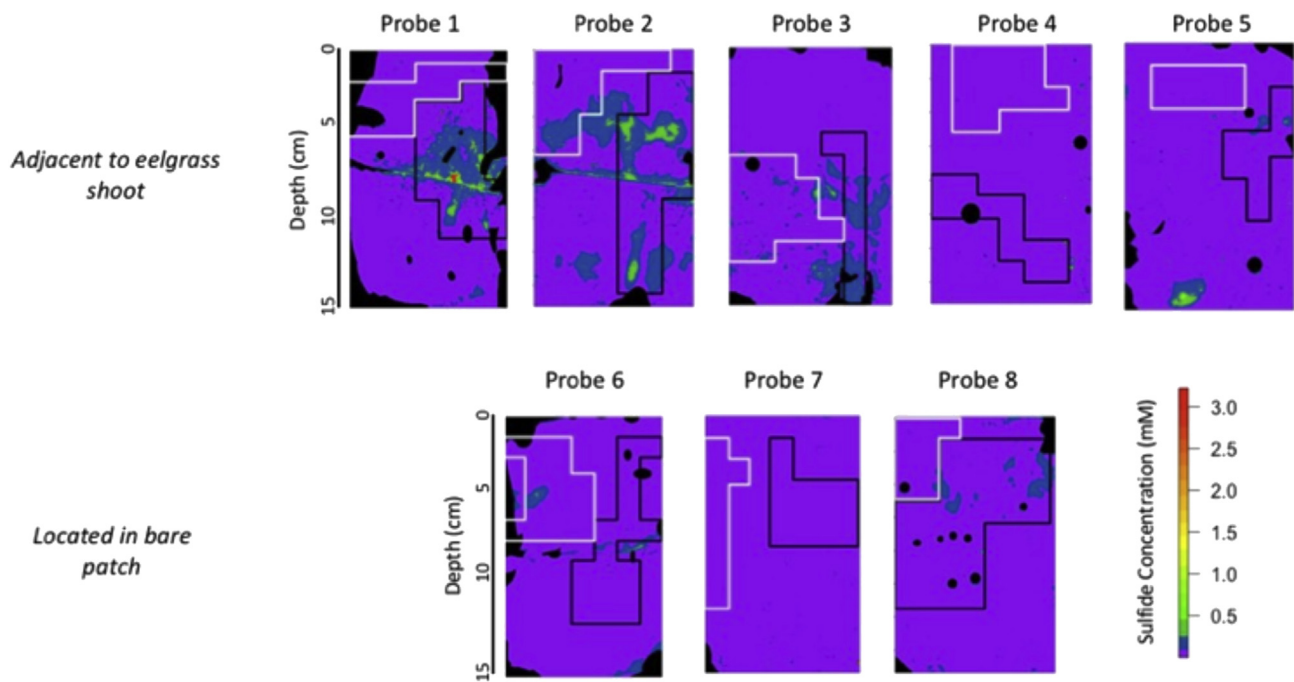
manipulated the quantity of leaf detritus and the presence or absence of eelgrass shoots in aquaria fitted with DGT or DET sensors. Sulfide concentrations in aquaria with and without live eelgrass shoots generally increased from week one to week two and stabilized during week three before decreasing during week four (ANCOVAR,  $F_{[1,7]} = 4.93$ ,  $p = 0.001$ ). Because differences between treatments first became apparent by week two, at which point, average sulfide levels changed little during the rest of the experiment, subsequent analyses were conducted using only week two data.

Aquaria containing live eelgrass shoots generally had higher average sulfide concentrations than the aquaria containing bare sediment even after considering the variation of added detritus (Fig. 5; including E1: ANCOVA,  $F_{[1,34]} = 9.828$ ,  $p = 0.003$ ; excluding E1: ANCOVA,  $F_{[1,30]} = 6.05$ ,  $p = 0.012$ ). Aquaria containing bare sediment had low pore-water sulfide concentrations with a mean sulfide concentration of  $0.13 \text{ mM}$ , whereas, the mean sulfide concentration for aquaria containing eelgrass was  $0.40 \text{ mM}$  (Fig. 5). Little difference was seen in mean Fe(II) concentrations between aquaria containing bare sediment and aquaria with live eelgrass, where the mean concentrations were  $80.3 \mu\text{M}$  and  $80.7 \mu\text{M}$ , respectively (Fig. 5). However, some DETs became contaminated with sediment, leading to the loss of some Fe(II) concentration data.

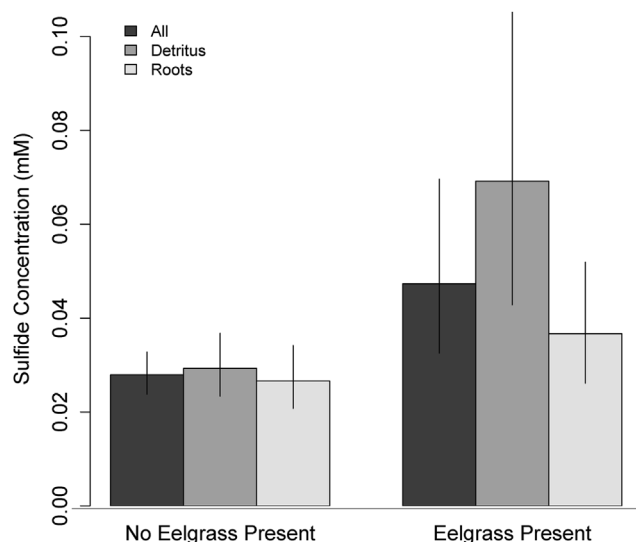
#### 3.3. Detritus additions

In some treatments, the eelgrass root zone extended deep enough to affect sulfide concentration at the 4-cm depth detritus addition locations. Here, we consider the effects of detritus only at 11-cm depth to minimize possible root zone effects. The addition of detritus yielded higher sulfide concentrations compared to sediment from the same depth without added detritus (Fig. 6; including E1: paired *t*-test,  $t = 2.69$ ,  $df = 9$ ,  $p = 0.025$ ; excluding E1: paired *t*-test,  $t = 2.90$ ,  $df = 8$ ,  $p = 0.019$ ). Using a pairwise *t*-test to compare the different detritus additions, the greatest increase in sulfide concentrations was seen between the 0 and 8-piece treatments (pairwise *t*-test,  $p < 0.001$ ). Fe(II) concentrations were similar in the detritus locations ( $82.73 \mu\text{M} \pm 10.25$  95% C.I.) compared to the whole ( $80.52 \mu\text{M} \pm 8.045$  95% C.I.,





**Fig. 3.** Field DGT probes analyzed at a 1-mm scale. White outlines indicate the combined grid cell areas where root tips were located. Roots were found in both bare patches and adjacent to eelgrass shoots. Black outlines indicate the combined grid cell areas where detritus was found. Probes 1, 2, 3, 4, and 5 were located adjacent to an eelgrass shoot, whereas probes 6, 7, and 8 were located in a bare patch.



**Fig. 4.** Analysis of field data based on the presence or absence of eelgrass. *All* refers to the mean sulfide concentration of the entire DGT sensor. *Detritus* refers to the mean sulfide concentration of areas containing detritus. *Roots* refers to the mean sulfide concentration of areas containing a root zone. Roots were found in both bare patches and adjacent to eelgrass shoots. Error bars represent 95% CI.

ANOVA,  $F_{[1,6]} = 2.522$ ,  $p = 0.16$ ).

### 3.4. Eelgrass shoot and rhizosphere presence

Eelgrass shoots grew during the four-week experiment gaining an average of 13.7 cm ( $\pm 6.1$  cm, s.d.) in blade length. Below the sediment surface, sulfide concentration around the rhizosphere was an order of magnitude lower than the adjacent non-root zone with a mean of 0.048 mM for the rhizosphere and a mean of 0.48 mM for the non-root zone within the same depth range (Fig. 7). The presence of roots

reduced sulfide concentrations compared to areas where no roots were present (*including E1*: paired  $t$ -test,  $t = 3.15$ ,  $df = 4$ ,  $p = 0.035$ ; *excluding E1*: paired  $t$ -test,  $t = 2.62$ ,  $df = 3$ ,  $p = 0.079$ ). The average Fe (II) concentration was slightly lower in root zones compared to non-root zones at the same depth, but this pattern was not statistically significant (paired  $t$ -test,  $t = -2.09$ ,  $df = 4$ ,  $p$ -value = 0.105).

### 3.5. Depth analysis

The depths at which sulfide concentrations were highest were typically between 8 and 14 cm, though some aquaria (E1 and E8) contained high sulfide concentrations throughout the entire depth range. Typically, deeper sediment possessed higher sulfide concentrations than surface sediment (Fig. 8; *including E1*: paired  $t$ -test,  $t = 3.52$ ,  $df = 19$ ,  $p = 0.002$ ; *excluding E1*: paired  $t$ -test,  $t = 3.11$ ,  $df = 17$ ,  $p = 0.006$ ). Notable peaks in Fe(II) concentration occurred at a depth of 5–7 cm in aquaria, slightly shallower than the peaks in sulfide concentration (Fig. 8).

## 4. Discussion

We determined that sulfide concentrations were highest in sediment surrounding leaf detritus in both the field and in our manipulative laboratory experiment, consistent with previous work by Boschker et al. (2000), and that sulfide concentration increased with increasing mass of added detritus. Furthermore, the presence of eelgrass shoots was found to simultaneously increase sulfide concentration away from the root zone (regardless of detritus presence or absence) and decrease concentration nearer the root zone. The enhancement of pore-water sulfide in sediment inhabited by eelgrass has been attributed to an input of dissolved organic carbon from roots (Wetzel and Penhale, 1979; Blaabjerg and Finster, 1998; Holmer and Bondgaard, 2001). Both types of organic matter (dissolved and particulate) contribute to the variation in sulfide distribution and concentration among and within aquaria. The drop in sulfide near roots is likely due to the release of oxygen from root tips (Frederiksen and Glud, 2006).

Our observation of higher average sulfide concentrations in the

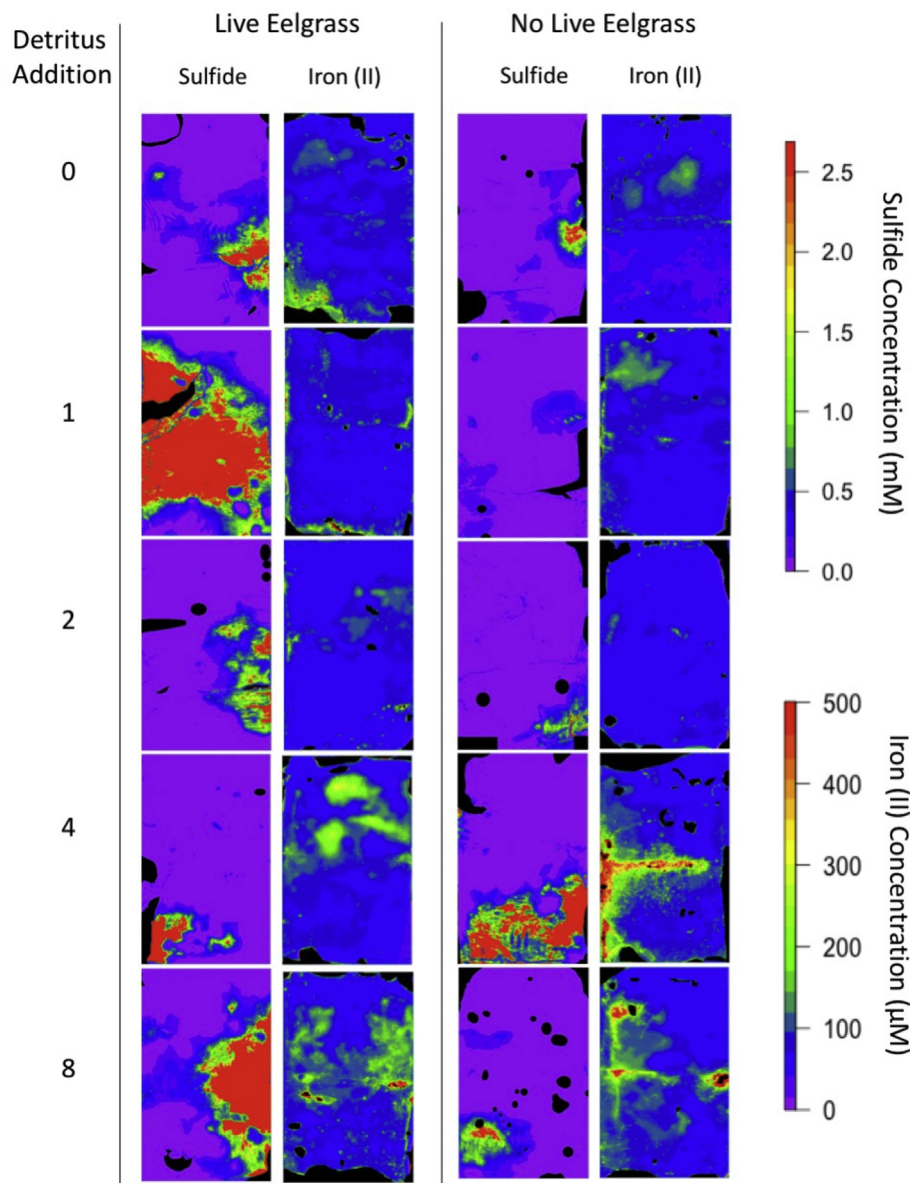


Fig. 5. DGT/DET images for sulfide and iron (II) concentrations from week two of laboratory experiment. Detritus addition refers to the number of detritus leaves added to the sediment.

presence of eelgrass shoots corroborates previous studies which indicate higher pore-water sulfide concentrations are found in eelgrass inhabited sediments (Holmer and Nielson, 1997; Blaabjerg and Finster, 1998; Blaabjerg et al., 1998) despite the risk of sulfide toxicity to eelgrass (Bagarinao, 1992; Holmer and Bondgaard, 2001; Koch and Erksine, 2001; Pedersen et al., 2004; Borum et al., 2005, 2014; Korhonen et al., 2012; Lamers et al., 2013). Furthermore, sulfate reduction rates have been found to be up to 20 times greater in sediments containing *Z. marina* roots and rhizomes than in sediments where the roots and rhizomes were removed (Blaabjerg and Finster, 1998; Blaabjerg et al., 1998; Welsh, 2000). Likewise, sulfate reduction has been found to be higher in vegetated than unvegetated sediments in *Zostera noltii* meadows (Nielsen et al., 2001). In the current study, the difference in sulfide concentrations between eelgrass-inhabited sediments and unvegetated sediments in the field study was not as great as seen in the laboratory study. This could be due to reduced pore-water exchange in the aquaria compared to the field site, which is exposed to greater tidal current velocities and periodic subaerial exposure (Padilla Bay NERR, 2011). Also, we sampled the field site during one low tide in

July. It is likely that sulfide concentrations would differ at other sites sampled at different times of the year. Nevertheless, eelgrass shoots in both the field and laboratory increased sulfide concentration within surrounding sediments which suggests that eelgrass presence somehow stimulates the activity of sulfate reducing bacteria (Jorgensen and Bak, 1991; Cifuentes et al., 2000; Welsh, 2000; Smith et al., 2004; Jensen et al., 2007).

The role of eelgrass detritus in stimulating sulfide production has also been examined in previous work. A similar pattern was observed when ground *Posidonia australis* leaf material was added homogeneously to *P. australis* meadows which led to more reducing sediments (Fraser et al., 2016). The addition of sucrose in *Z. marina* and *Cymodocea nodosa* meadows also increased sulfide concentrations (Terrados et al., 1999). Organic enrichment leading to high sulfide concentrations also has been linked to the massive die-off of the tropical seagrass *Thalassia testudinum* (Carlson et al., 1994) and to the loss of *Z. marina* in Puget Sound (Elliot et al., 2006). However, as shown in this study, the effects of increased sulfide concentrations appear to be localized with high concentrations immediately in the vicinity of leaf

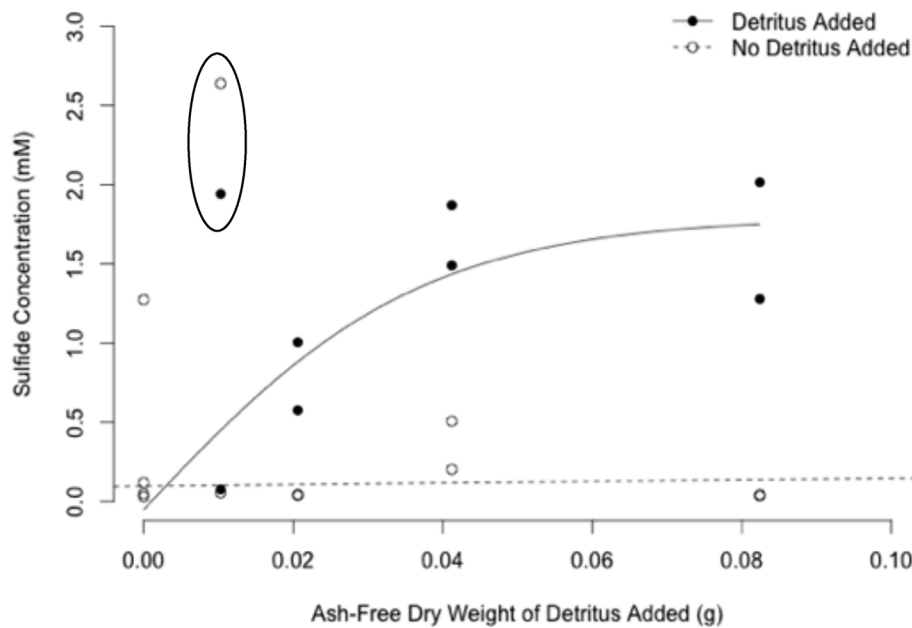


Fig. 6. Average sulfide concentrations for locations with and without detritus leaves added at 11 cm depth versus mass of detritus added. The regression lines show the relationship between sulfide concentration and mass of detritus added when not including aquarium E1. The circled points indicate the values obtained from aquarium E1.

detritus. Previous studies where organic matter was intentionally added to seagrass meadows (Terrados et al., 1999; Fraser et al., 2016) did not indicate that sulfide concentrations increased primarily around the site of organic matter addition. It is likely, though, that the researchers were unable to observe this pattern using their measurement technique (redox potential). In addition, we did not observe elevated Fe(II) in this study except for the aquaria with eight added detritus leaves. Here, it is likely that the rate of iron reduction in the vicinity of eelgrass detritus exceeded the rate of sorption and iron sulfide formation, allowing for measurable concentrations of Fe(II) to remain in solution. This indicates that the addition of organic matter initiates as well as accelerates anaerobic metabolism.

In the field, the presence of a root zone contributed to lower mean sulfide concentrations compared to sediment without a root zone. Furthermore, in the laboratory study, the root zone had approximately an order of magnitude lower sulfide concentrations than the non-root zone. Even though the oxic zone around *Z. marina* root tips only extends

~2–3 mm beyond the root tip, clusters of root tips can create a much larger area of dissolved oxygen extending up to 8 mm along the root (Frederiksen and Glud, 2006). High oxygen production around the root zone is also found in other seagrasses, such as *Zostera muelleri* (Koren et al., 2015). Oxygen leakage from the roots, however, is highly localized and maintains its largest impact near the root tip (Koren et al., 2015). In multiple eelgrass aquaria, the impact on sulfide concentration from clustered root tips was noticeable. For instance, the top corner of aquarium E1, where the eelgrass shoot was planted, had lower concentrations of sulfide and Fe(II) each week compared to other parts of the sediment. And, in E8, high sulfide concentrations were found throughout the DGT except in the location where eelgrass roots were observed.

Sulfate reduction typically occurs at depth in the sediment due to microbial preference for other more energetically favorable electron acceptors closer to the surface (Burdige, 2006; Mitsch and Gosselink, 2007). In the aquarium sediments, the highest sulfide concentrations

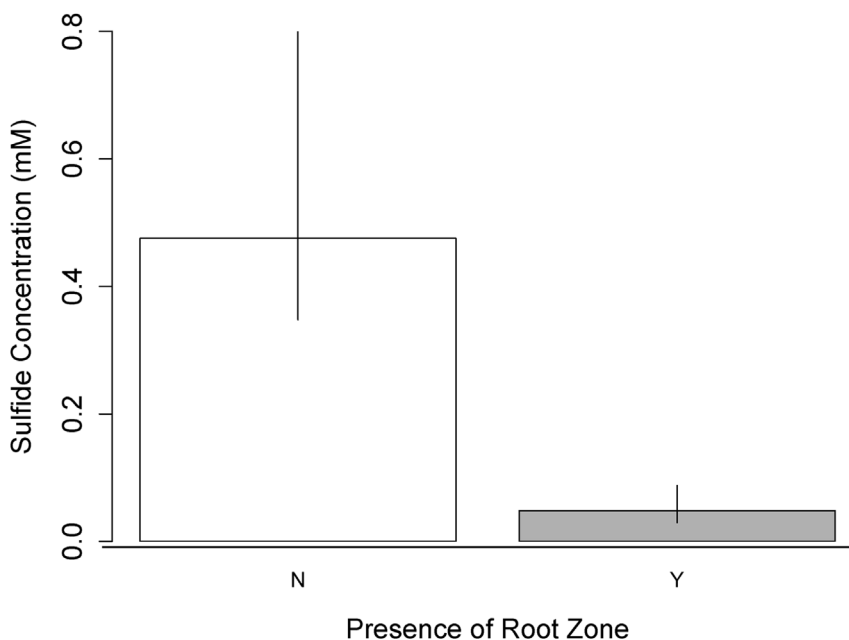
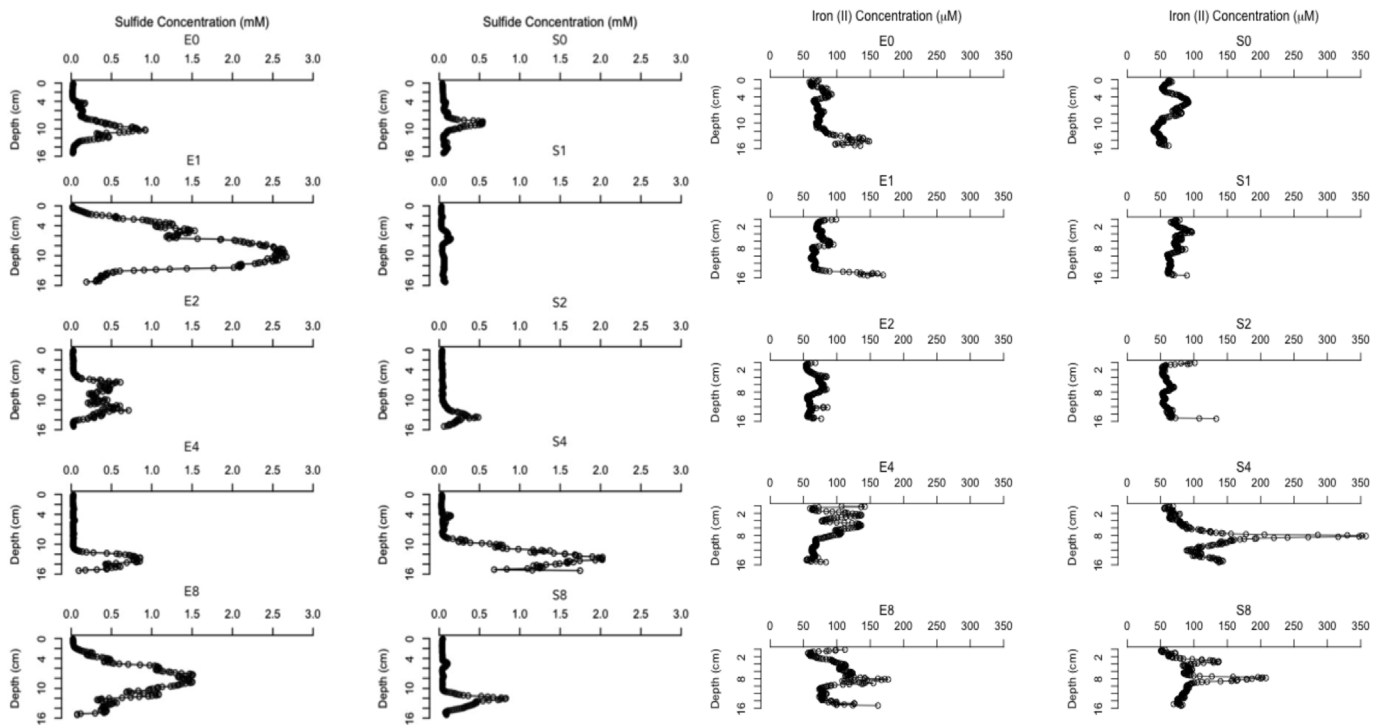


Fig. 7. Comparison of sulfide concentrations for eelgrass aquaria in locations with and without roots. The root zone was defined as the area where the majority of root tips were located. Sulfide concentrations from the no root zone category were determined by using an area of the same size as the root zone on the opposite side of the DGT. Error bars indicate 95% CI (n = 5) after back transforming data.



**Fig. 8.** Analysis of sulfide and iron concentrations by depth for all aquaria. The x-axis (A) represents horizontally averaged sulfide concentrations in mM at 1 mm scales with the detritus hotspots included in calculations. The x-axis (B) represents horizontally averaged iron (II) concentrations in  $\mu\text{M}$  at 1 mm scales with the detritus hotspots included in calculations. E indicates aquaria with a live eelgrass shoot planted while S indicates aquaria with sediment only. The corresponding number (0, 1, 2, 4, or 8) indicates the number of eelgrass detritus leaves added at 4 cm and 11 cm depth.

typically occurred from approximately 8 to 12 cm in depth and the highest Fe(II) concentrations typically occurred from 4 to 7 cm in depth, which is fairly typical of coastal sediments (Burdige, 2006). The wide depth range of sulfide in vegetated aquaria seems typical for eelgrass-inhabited sediments especially during summer months when sulfide can be found even near the sediment surface (Frederiksen et al., 2006). A wide spatial variation in sulfide concentrations has been noted in other seagrass species, such as *T. testudinum*, as well (Chambers et al., 2001). Additionally, a study on *Zostera noltii* inhabited sediments suggested that the top few centimeters of vegetated sediment yielded nearly twice the sulfide production compared to unvegetated sediments (Isaken and Finster, 1996). This further confirms that the spatial distribution of sulfide in seagrass vegetated sediments is far wider than in unvegetated sediments. Sulfate reducing bacteria must also utilize organic matter from multiple sources in addition to eelgrass leaf detritus.

Sulfide concentration also varied with Fe(II) concentration with lower sulfide concentrations corresponding to higher Fe(II) concentrations. Because pore water Fe(II) precipitates with free sulfide to form iron sulfide, reducing the concentration of free sulfide (Jørgensen, 1977), we infer that pore water sulfide concentrations were low at greater depth in the sediment due to its precipitation. Concentrations of reactive (ammonium oxalate extracted) iron oxides are highest in the top 2–4 cm in Padilla Bay sediments (unpublished data). This is consistent with the observation that Fe(II) concentrations were higher in some of the tanks surrounding implanted eelgrass detritus at 4 cm depth, where we would expect to find reducible Fe(OH)x, but generally low near the deep implanted eelgrass detritus. Similarly, Robertson et al. (2008) observed that areas with high sulfide concentration in sediment corresponded to areas of low Fe(II) concentration, which was similarly attributed to the production of iron sulfide.

The influence of eelgrass leaf detritus on sulfide concentrations has the potential to extend beyond the eelgrass meadow as seagrasses have been found to be important sources of organic carbon in habitats as distant as the deep sea (Duarte and Krause-Jensen, 2017; Heck et al.,

2008). While nearly half of the carbon produced in eelgrass meadows is remineralized within the meadow itself (Duarte and Krause-Jensen, 2017), a large portion of dissolved organic carbon (DOC) and particulate organic carbon (POC) are exported from seagrass meadows (Duarte and Krause-Jensen, 2017; Barrón et al., 2014; Duarte and Cebrián, 1996). Of the 120 Tg C year<sup>-1</sup> exported from seagrass meadows, 87 Tg C year<sup>-1</sup> is derived from seagrass detritus (Duarte and Krause-Jensen, 2017) while the remaining 33 Tg C year<sup>-1</sup> is derived from seagrass DOC (Barrón et al., 2014). As eelgrass leaf detritus was found to be an important contributor to sulfide production and rafts of eelgrass leaves can be transported large distances (Duarte and Krause-Jensen, 2017; Heck et al., 2008), eelgrass can potentially enhance sulfate reduction in distant habitats.

The results presented here confirm the importance of eelgrass in influencing sediment biogeochemistry. Sediment biogeochemical cycling in vegetated sediments could not be explained by only one factor (i.e., rhizosphere or detritus) but instead resulted from multiple processes. This notion challenges previous findings that seagrasses influenced chemical cycling either through releasing oxygen through roots (Wetzel and Penhale, 1979; Blaabjerg and Finster, 1998; Holmer and Bondgaard, 2001) or through the addition of organic matter (Boscher et al., 2000), not both. We observed that the presence of eelgrass can simultaneously increase the sulfide concentrations away from the root zone, in part through the provision of leaf detritus, and decrease concentrations within the root zone.

## Acknowledgements

We thank M. Adamczyk, T. Tran, and K. Wright for help in the lab and field. We also thank M. Rearick for guidance on DGT preparation. This research was supported by a grant from Washington Sea Grant (R/HCE-7) to D.H. Shull and S. Yang and an NSF REU fellowship to L. Tripp.



## References

- Bagarinao, T., 1992. Review: sulfide as an environmental factor and toxicant: tolerance and adaptations in aquatic organisms. *Aquat. Toxicol.* 24, 21–62.
- Barrón, C., Apostolaki, E.T., Duarte, C.M., 2014. Dissolved organic carbon fluxes by seagrass meadows and macroalgal beds. *Front. Mar. Sci.* 1, 42.
- Blaabjerg, V., Finster, K., 1998. Sulphate reduction associated with roots and rhizomes of the marine macrophyte *Zostera marina*. *Aquat. Microb. Ecol.* 15, 311–314.
- Blaabjerg, V., Mouritsen, K.N., Finster, K., 1998. Diel cycles of sulphate reduction rates in sediments of a *Zostera marina* bed (Denmark). *Aquat. Microb. Ecol.* 15, 97–102.
- Borum, J., Pedersen, O., Greve, T.M., Frankovich, T.A., Ziemann, J.C., Fourqurean, J.W., Madden, C.J., 2005. The potential role of plant oxygen and sulphide dynamics in die-off events of the tropical seagrass, *Thalassia testudinum*. *J. Ecol.* 93, 148–158.
- Borum, J., Raun, A.L., Hasler-Sheetal, H., Pedersen, M.O., Pedersen, O., Holmer, M., 2014. Eelgrass fairy rings: sulfide as inhibiting agent. *Mar. Biol.* 161, 351–358.
- Boschker, H.T.S., Wielemaaker, B.E.M., Schaub, A., Holmer, M., 2000. Limited coupling of macrophyte production and bacterial carbon cycling in the sediments of *Zostera* spp. meadows. *Mar. Ecol. Prog. Ser.* 203, 181–189.
- Burdige, D.J., 2006. *Geochemistry of Marine Sediments*. Princeton University Press, Princeton.
- Carlson Jr., P.R., Yarbrow, L.A., Barber, T.R., 1994. Relationship of sediment sulfide to mortality of *Thalassia testudinum* in Florida Bay. *Bull. Mar. Sci.* 54 (3), 733–746.
- Cesbron, F., Metzger, E., Launeau, P., Deflandre, B., Delgard, M., Thibault de Chanvalon, A., Geslin, E., Anschutz, P., Jezequel, D., 2014. Simultaneous 2D imaging of dissolved iron and reactive phosphorus in sediment porewaters by thin-film and hyperspectral methods. *Environ. Sci. Technol.* 48 (5), 2816–2826.
- Chambers, R.M., Fourqurean, J.W., Macko, S.A., Hoppenot, R., 2001. Biogeochemical effects of iron availability on primary producers in a shallow marine carbonate environment. *Limnol. Oceanogr.* 46, 1278–1286.
- Cifuentes, A., Antón, J., Benlloch, S., Donnelly, A., Herbert, R.A., Rodríguez-Valera, F., 2000. Prokaryotic diversity in *Zostera noltii*-colonized marine sediments. *Appl. Environ. Microbiol.* 66, 1715–1719.
- Deborde, J., Abril, G., Mouret, A., Jezequel, D., Thouzeau, G., Clavier, J., Bachelet, G., Anschutz, P., 2008. Effects of seasonal dynamics in a *Zostera noltii* meadow on phosphorus and iron cycles in a tidal mudflat (Arcachon Bay, France). *Mar. Ecol. Prog. Ser.* 355, 59–71.
- Dooley, F.D., Wyllie-Echeverria, S., Roth, M.B., Ward, P.D., 2013. Tolerance and response of *Zostera marina* seedlings to hydrogen sulfide. *Aquat. Bot.* 105, 7–10.
- Dowty, P., Reeves, B., Berry, H., Wyllie-Echeverria, S., Mumford, T., Sewell, A., Milos, P., Wright, R., 2005. Puget Sound Submerged Vegetation Monitoring Project 2003–2004 Monitoring Project. Nearshore Habitat Program, Aquatic Resources Division, Washington State Department of Natural Resources, Olympia, WA.
- Duarte, C.M., Cebrián, J., 1996. The fate of marine autotrophic production. *Limnol. Oceanogr.* 41, 1758–1766.
- Duarte, C.M., Krause-Jensen, D., 2017. Export from seagrass meadows contributes to marine carbon sequestration. *Front. Mar. Sci.* 4, 13.
- Elliot, J.K., Spear, E., Wyllie-Echeverria, S., 2006. Mats of *Beggiatoa* bacteria reveal that organic pollution from lumber mills inhibits growth of *Zostera marina*. *Mar. Ecol. Prog. Ser.* 325, 372–380.
- Fraser, M.W., Statton, J., Hovey, R.K., Laverock, B., Kendrick, G.A., 2016. Seagrass derived organic matter influences biogeochemistry, microbial communities and seedling biomass partitioning in seagrass sediments. *Plant Soil* 400, 133–146.
- Frederiksen, M.S., Glud, R.N., 2006. Oxygen dynamics in the rhizosphere of *Zostera marina*: a two-dimensional planar optode study. *Limnol. Oceanogr.* 51 (2), 1072–1083.
- Frederiksen, M.S., Holmer, M., Borum, J., Kennedy, H., 2006. Temporal and spatial variation of sulfide invasion in eelgrass (*Zostera marina*) as reflected by its sulfur isotopic composition. *Limnol. Oceanogr.* 51 (5), 2308–2318.
- Harrison, P.G., Mann, K.H., 1975. Detritus formation from eelgrass (*Zostera marina* L.): the relative effects of fragmentation, leaching, and decay. *Limnol. Oceanogr.* 20 (6), 924–934.
- Heck Jr., K.L., Carruthers, T.J.B., Duarte, C.M., Hughes, A.R., Kendrick, G., Orth, R.J., Williams, S.W., 2008. Trophic transfers from seagrass meadows subsidize diverse marine and terrestrial consumers. *Ecosystems* 11, 1198–1210.
- Holmer, M., Bondgaard, E.J., 2001. Photosynthetic and growth response of eelgrass to low oxygen and high sulfide concentrations during hypoxic events. *Aquat. Bot.* 70, 29–38.
- Holmer, M., Frederiksen, M.S., Møllegaard, H., 2005. Sulfur accumulation in eelgrass (*Zostera marina*) and effect of sulfur on eelgrass growth. *Aquat. Bot.* 81 (4), 367–379.
- Holmer, M., Nielsen, S.L., 1997. Sediment sulfur dynamics related to biomass density patterns in *Zostera marina* (eelgrass) beds. *Mar. Ecol. Prog. Ser.* 146, 163–171.
- Isaksen, M.F., Finster, K., 1996. Sulphate reduction in the root zone of the seagrass *Zostera noltii* on the intertidal flats of a coastal lagoon (Arcachon, France). *Mar. Ecol. Prog. Ser.* 137, 187–194.
- Jensen, S.I., Kühl, M., Priemé, A., 2007. Different bacterial communities associated with the roots and bulk sediment of the seagrass *Zostera marina*. *FEMS Microb. Ecol.* 62, 108–117.
- Jørgensen, B.B., 1977. The sulfur cycle of a coastal marine sediment (Limfjorden, Denmark). *Limnol. Oceanogr.* 22, 814–832.
- Jørgensen, B.B., Bak, F., 1991. Pathways and microbiology of thiosulfate transformations and sulfate reduction in a marine sediment (Kattegat, Denmark). *Appl. Environ. Microbiol.* 57, 847–857.
- Koch, M.S., Erskine, J.M., 2001. Sulfide as a phytotoxin to the tropical seagrass *Thalassia testudinum*: interactions with light, salinity and temperature. *J. Exp. Mar. Biol. Ecol.* 266, 81–95.
- Koren, K., Brodersen, K.E., Jakobsen, S.L., Kühl, M., 2015. Optical sensor nanoparticles in artificial sediments – a new tool to visualize O<sub>2</sub> dynamics around the rhizome and roots of seagrasses. *Environ. Sci. Technol.* 49 (4), 2286–2292.
- Korhonen, L.K., Macías-Carranza, V., Abdala, R., Figueroa, F.L., Cabello-Pasini, A., 2012. Effects of sulfide concentration, pH, and anoxia on photosynthesis and respiration of *Zostera marina*. *Cienc. Mar.* 38 (4), 625–633.
- Lamers, L.P.M., Govers, L.L., Janssen, I.C.J.M., Geurts, J.J.M., Van der Welle, M.E.W., Van Katwijk, M.M., Van der Heide, T., Roelofs, J.G.M., Smolders, A.J.P., 2013. Sulfide as a soil phytotoxin – a review. *Front. Plant Sci.* 4, 1–14.
- Mitsch, W.J., Gosselink, J.G., 2007. *Wetlands*. John Wiley & Sons Inc, Hoboken.
- Nearshore Habitat Program, 2015. Puget Sound Submerged Vegetation Monitoring Program: 2010–2013 Report. DNR Nearshore Habitat Program, Olympia, WA.
- Nielsen, L.B., Finster, K., Welsh, D.T., Donnelly, A., Herbert, R.A., de Wit, R., Lomstein, B.A., 2001. Sulphate reduction and nitrogen fixation rates associated with roots, rhizomes and sediments from *Zostera noltii* and *Spartina maritima* meadows. *Environ. Microbiol.* 3 (1), 63–71.
- Padilla Bay NERR, 2011. Padilla Bay National Marine Estuarine Research Reserve Management Plan. Padilla Bay National Estuarine Research Reserve, Mount Vernon, WA, pp. 181.
- Pages, A., Welsh, D.T., Robertson, D., Panther, J.G., Schäfer, J., Tomlinson, R.B., Teasdale, P.R., 2012. Diurnal shifts in co-distributions of sulfide and iron(II) and profiles of phosphate and ammonium in the rhizosphere of *Zostera capricorni*. *Estuar. Coast Shelf Sci.* 115, 282–290.
- Pedersen, O., Binzer, T., Borum, J., 2004. Sulphide intrusion in eelgrass (*Zostera marina* L.). *Plant Cell Environ.* 27, 595–602.
- Pollard, P.C., Moriarty, D.J.W., 1991. Organic carbon decomposition, primary and bacterial productivity, and sulphate reduction, in tropical seagrass beds of the Gulf of Carpenteria, Australia. *Mar. Ecol. Prog. Ser.* 69, 149–159.
- R Core Team, 2016. R: a Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>.
- Rasband, W.S., 2015. ImageJ. U. S. National Institutes of Health, Bethesda, MD. <http://imagej.nih.gov/ij/>.
- Rearick, M.S., 2004. In Situ Measurement of Sulfide in Natural Waters. Master's Thesis. University of Maryland, College Park, MD.
- Robertson, D., Teasdale, P.R., Welsh, D.T., 2008. A novel gel-based technique for the high resolution, two-dimensional determination of iron (II) and sulfide in sediment. *Limnol. Oceanogr.* 53, 502–512.
- Short, F.T., Duarte, C.M., 2001. Chapter 8: methods for the measurement of seagrass growth and production. In: Short, F.T., Coles, R.G. (Eds.), *Global Seagrass Research Methods*. Elsevier, pp. 155–182.
- Smith, A.C., Kostka, J.E., Devereux, R., Yates, D.F., 2004. Seasonal composition and activity of sulfate-reducing prokaryotic communities in seagrass bed sediments. *Aquat. Microb. Ecol.* 37, 183–195.
- Terrados, J., Duarte, C.M., Kamp-Nielsen, L., Agawin, N.S.R., Gacia, E., Lacap, D., Fortes, M.D., Borum, J., Lubanski, M., Greve, T., 1999. Are seagrass growth and survival constrained by the reducing conditions of the sediment? *Aquat. Bot.* 65, 175–197.
- Walser, A., 2014. A Study of Pore-water Sulfide and Eelgrass (*Zostera Japonica* and *Zostera Marina*) in Padilla Bay, Washington. Master's Thesis. Western Washington University, Bellingham, WA.
- Welsh, D.T., 2000. Nitrogen fixation in seagrass meadows: regulation, plant–bacteria interactions and significance to primary productivity. *Ecol. Lett.* 3, 58–71.
- Wetzel, R.G., Penhale, P.A., 1979. Transport of carbon and excretion of dissolved organic carbon by leaves and roots/rhizomes in seagrasses and their epiphytes. *Aquat. Bot.* 6, 149–158.
- Zhang, H., Davison, W., 1995. Performance characteristics of diffusion gradients in thin films for the in-situ measurement of trace metals in aqueous solution. *Anal. Chem.* 67, 3391–3400.